UltraDrill: A Distillation and Optimization Framework for Edge Ultrasound Lesion Classification

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Abstract

Background and Objectives: Ultrasound lesion classifiers are usually trained and reported on annotation-derived region-of-interest (ROI) crops, but mobile use depends on imperfect localization. A deployment-oriented classifier should therefore be selected not only by clean-ROI accuracy, but also by the performance loss caused by plausible ROI errors. **Methods:** We propose UltraDrill, a distillation and optimization framework for edge ultrasound lesion classification. UltraDrill contains four linked modules: ROI preprocessing with localization-stress construction, high-capacity teacher optimization and selection, EfficientFormer student distillation, and edge post-processing with Android export verification. The ROI-stress module evaluates ground-truth boxes, synthetic box noise, detector-derived boxes, and full-image inputs before and after distillation, so teacher selection and mobile student verification use the same stress test. Experiments use TN5000 thyroid nodules as the primary benchmark, with BUSI breast and AUL liver datasets used as additional public in-domain benchmarks and for multi-organ boundary analysis. **Results:** On TN5000, UL-TransXNet reaches AUC 0.9537 and balanced accuracy 0.8865, slightly above ConvNeXt-Tiny under the fixed protocol. Full-factorial ablation identifies TransXNet-MUDD+DA as the strongest teacher configuration (AUC 0.9579, balanced accuracy 0.8901), while coordinate-aware refinement is dataset-dependent. Under the unified localization-stress protocol, the MUDD+DA teacher obtains GT-ROI

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AUC 0.9623, 20% box-noise AUC 0.9463, and detector-ROI AUC 0.9497. The distilled EfficientFormer-L1+KD student obtains GT-ROI AUC 0.9585, 20% box-noise AUC 0.9529, and detector-ROI AUC 0.9534, showing that distillation does not amplify localization sensitivity. The EfficientFormer-L1+ECA+KD Android artifact reaches 0.9552 TN5000 AUC with 31.3/56.5 ms hot latency on Xiaomi and Samsung devices. Joint multi-organ training is feasible, but leave-one-domain-out transfer remains weak. **Conclusions:** UltraDrill turns ROI localization stress into a framework-level criterion for teacher selection, student distillation, and edge verification. The resulting evidence supports a bounded edge-ultrasound classification workflow: optimize the teacher under localization stress, distill compact students, apply fixed post-processing, and verify both robustness and real-device latency before claiming deployment feasibility.

Keywords: Ultrasound image classification, Edge AI, Knowledge distillation, Model optimization, Localization robustness, Mobile inference

1. Introduction

Ultrasound is widely used for screening, follow-up, and diagnostic support because it is inexpensive, radiation-free, portable, and suitable for repeated acquisition. These advantages also create a difficult computer-aided diagnosis setting. Speckle noise, weak lesion boundaries, low contrast, operator dependency, and device-dependent appearance shifts can obscure benign–malignant cues. Deep learning has improved ultrasound classification, but practical lesion classifiers still face a specific deployment gap: many models are optimized and reported on annotation-derived ROI crops, whereas mobile use must tolerate imperfect lesion localization.

This paper targets that gap from a computing-methods perspective through UltraDrill, a distillation and optimization framework for edge ultrasound lesion classification. Rather than presenting a new universal medical vision backbone, UltraDrill organizes the steps needed to build a mobile classifier from ROI data: preprocess and stress-test lesion crops, optimize and select a reliable high-capacity teacher, distill a compact student, and apply fixed post-processing and export checks before real-device measurement. In this framing, localization robustness is not an auxiliary diagnostic table; it is a framework-level criterion that connects teacher selection, student compression, and edge verification.

TN5000 thyroid nodules are used as the primary benchmark [1], while BUSI breast lesions [2] and the Annotated Ultrasound Liver (AUL) dataset [3] provide additional public in-domain benchmarks from different organs. These datasets differ in anatomy, acquisition appearance, annotation protocol, and class prevalence, so they should not be treated as automatic evidence of zero-shot cross-organ generalization. We use a TransXNet-family teacher [4] as a high-capacity ROI model and study three adaptations: coordinate-aware feature refinement inspired by coordinate attention [5], multi-scale dynamic dense reuse, and differential attention adapted from the Differential Transformer formulation [6]. We then distill EfficientFormer-L1 students [7] and measure exported artifacts on real Android devices.

The paper is organized around three framework questions. First, how should an edge-ultrasound classification pipeline preprocess ROI inputs and quantify localization stress? Second, which high-capacity teacher should be selected for distillation when clean-ROI accuracy and localization robustness are both considered? Third, after distillation and post-processing, does the exported student preserve the same robustness while satisfying real-device latency constraints?

The contributions are:

- We introduce UltraDrill, a modular edge-ultrasound classification framework that links ROI preprocessing, teacher optimization, student distillation, and deployment post-processing under one fixed reporting protocol.

- We instantiate the preprocessing module with a localization-stress protocol that evaluates GT boxes, synthetic box perturbations, detector-derived boxes, and full-image inputs under a fixed validation-derived operating point.

- We instantiate the teacher-optimization module with TransXNet-family MCA/MUDD/DA ablations, showing that MUDD and differential attention provide the most reliable TN5000 teacher configuration.

- We instantiate the distillation module with EfficientFormer-L1 KD students and show under the same localization-stress protocol that compression does not amplify ROI-error sensitivity.

- We instantiate the post-processing and edge-verification module with fixed calibration/threshold logic, ONNX Runtime export, and Xiaomi/Samsung Android measurements, while BUSI/AUL and LODO probes define the boundary of dataset generalization.

2. Related work

Deep learning is now central to medical image analysis [8], but ultrasound remains difficult because acquisition conditions, speckle, boundary ambiguity, and annotation quality vary substantially across datasets. Clinical reporting systems such as ACR TI-RADS for thyroid ultrasound [9] and BI-RADS for breast imaging [10] also emphasize morphology, margin, and contextual appearance rather than isolated texture alone. Earlier breast-ultrasound classification work already showed that ROI definition and feature design strongly affect benign–malignant discrimination [11]. More recent ultrasound studies have explored multi-view learning for tumor classification [12], segmentation-oriented architectures [13], lightweight real-time models [14, 15], dataset curation for BUSI [16], and ultrasound foundation-model pretraining [17]. These studies motivate the present focus on the interface between lesion localization, ROI classification, data quality, and computational efficiency rather than on accuracy alone.

Efficient vision backbones provide another relevant line of work. CNNs and modern hybrid models such as ConvNeXt, Swin Transformer, RepViT, EfficientFormer, and TransXNet show different trade-offs among parameters, MACs, GPU latency, and mobile latency [18, 19, 20, 7, 4]. In medical ultrasound, however, parameter count alone is not a sufficient deployment claim. Preprocessing, TTA, model export, runtime delegate choice, and device-specific kernels can dominate practical latency. This is why our evaluation separates the high-capacity TransXNet-family teacher from the distilled EfficientFormer student.

Finally, robustness and reporting transparency are important for clinical AI. Calibration errors can make probability outputs unreliable even when AUC is high [21], and imperfect medical annotations or localization errors can change downstream performance [22]. We therefore report validation-fixed selection, detector-derived ROI experiments, Android runtime measurements, and leave-one-domain-out probes. These are not add-on diagnostics; they define the claim boundary of the method.

3. Methods

3.1. UltraDrill framework overview

UltraDrill is organized as a four-module framework for edge ultrasound lesion classification. The first module is ROI preprocessing and localization-stress construction: annotation boxes, synthetic perturbations, detector boxes, and full-image controls define the inputs used to quantify how much classification changes when localization is imperfect. The second module is teacher optimization and selection: high-capacity TransXNet-family variants are trained and ablated, then selected by both clean-ROI performance and localization-stress behavior. The third module is student distillation and optimization: teacher logits supervise EfficientFormer-L1 students, optionally with lightweight channel attention, while the same ROI preprocessing and validation protocol are kept fixed. The fourth module is post-processing and edge verification: validation-derived thresholds, temperature scaling, TTA/checkpoint averaging choices, ONNX export, and Android runtime measurement are treated as part of the reported system rather than as informal deployment details. The sections below describe these modules in order.

3.2. Problem setting and ROI construction

Given an ultrasound image and a lesion region, the classifier predicts a binary label $y \in \{0, 1\}$, where 0 denotes benign and 1 denotes malignant. The model produces logits $z = f_\theta(x)$ and a malignant posterior

$$p(y = 1 \mid x) = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)}. \tag{1}$$

The main protocol constructs expanded lesion-centered ROI crops from annotation boxes, as summarized in Fig. 1. The box is expanded around the lesion center, clipped to the image boundary, resized to the model input size, and normalized. This oracle-ROI setting isolates classifier behavior and represents a controlled upper-bound input. We additionally evaluate detector-derived boxes to estimate practical loss when annotation boxes are unavailable.

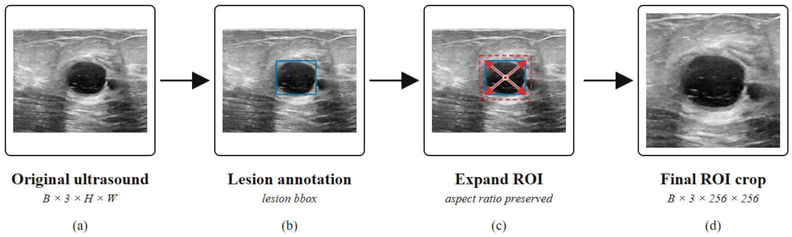

Figure 1: ROI input construction. The lesion annotation defines an initial box, which is expanded while preserving aspect ratio and then resized to the classifier input size.

3.3. Teacher optimization and selection

The teacher is a TransXNet-family hierarchical local–global backbone. It follows a four-stage configuration with stage depths $[4, 4, 15, 4]$, embedding dimensions $[48, 96, 224, 448]$, head numbers $[1, 2, 4, 8]$, and spatial-reduction ratios $[8, 4, 2, 1]$. For compatibility with the original implementation and the saved experimental checkpoints, the backbone is configured with a 1000-D trainable task projection before the binary head. This vector is not interpreted as ImageNet class probabilities; it is fine-tuned end-to-end as an intermediate representation and then passed to a 1000–512–2 MLP head.

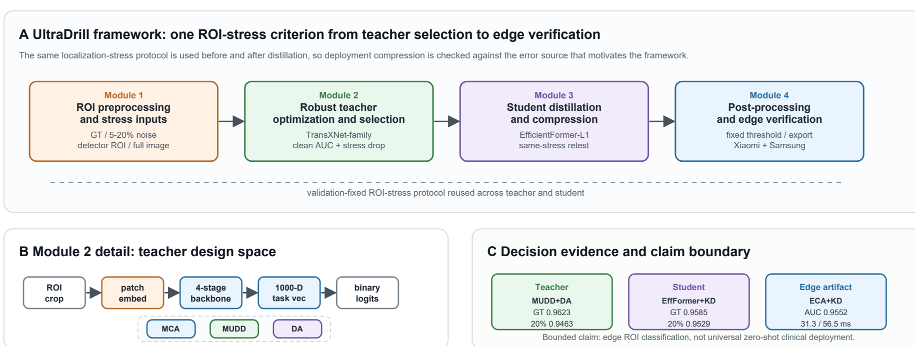

Figure 2: UltraDrill framework overview. (A) The four modules use the same localization-stress criterion from ROI preprocessing through teacher selection, student distillation, and edge verification. (B) The teacher-optimization module searches a TransXNet-family design space with coordinate-aware refinement, dynamic dense reuse, and differential attention. (C) The reported outputs and boundary statements are tied to teacher robustness, student robustness, and two-device Android evidence.

We study three modular adaptations. Multi-scale coordinate attention (MCA) is a coordinate-aware refinement option for spatially selective feature recalibration. Multiway dynamic dense connection (MUDD) is inserted before the token-mixer residual branch in blocks from stage 2 onward. Within each stage, it stores at most four previous block outputs and resets the cache at stage boundaries. For a current feature map $x_l \in \mathbb{R}^{B\times C\times H\times W}$, MUDD first splits channels into $P = 4$ equal paths $x_{l,p}$. A lightweight global gate produces a path-mixing matrix

$$W_l = \mathrm{softmax}\left(\mathrm{Conv}_{1\times 1}\left(\phi(\mathrm{Conv}_{1\times 1}(\mathrm{GAP}(x_l)))\right)\right) \in \mathbb{R}^{B\times P\times P}, \tag{2}$$

where ϕ is GELU. Each path is transformed by a depthwise 3×3 convolution and a pointwise 1×1 convolution, then receives spatially aligned same-path features from previous blocks:

$$\tilde{x}_{l,p} = T_p(x_{l,p}) + \sum_{j<l} W_l[:, p, j \bmod P]\,\mathrm{Resize}(x_{j,p}; H, W). \tag{3}$$

The path outputs are concatenated to recover C channels and are then processed by the original TransXNet block. Thus MUDD is a dynamic feature-reuse connection, not a separate classifier head. Differential attention computes two attention maps and subtracts the second from the first:

$$A_{\mathrm{diff}} = \mathrm{softmax}(Q_1K_1^\top/\sqrt{d}) - \lambda \mathrm{softmax}(Q_2K_2^\top/\sqrt{d}), \tag{4}$$

where λ is learned. In this ROI-ultrasound setting, the module is interpreted as a way to suppress common background or speckle-like responses while sharpening lesion-related relational contrast. We do not claim these modules as new independent primitives; the contribution is their controlled reorganization and evaluation for ROI-robust ultrasound classification.

3.4. Student distillation and edge post-processing

The TransXNet-family teacher is parameter/MAC-efficient relative to several large baselines, but it is not optimized for real-device latency. UltraDrill therefore trains EfficientFormer-L1 students with teacher guidance. This separates two roles: the TransXNet-family model provides an accuracy-oriented teacher and design analysis, whereas the EfficientFormer student is the edge candidate. For student training, teacher logits z_t are generated from the validation-selected TransXNet-family teacher under the same TN5000

ROI preprocessing. The student logits z_s are optimized with a standard supervised and distillation objective,

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\mathrm{CE}}(y, \sigma(z_s)) + \alpha\tau^2 \mathrm{KL}\left(\sigma(z_t/\tau) \,\|\, \sigma(z_s/\tau)\right), \tag{5}$$

where τ is the distillation temperature and α controls the supervised–teacher trade-off. In the reported TN5000 mobile runs, $\alpha = 0.45$ and $\tau = 3.0$. Students are trained for up to 70 epochs with early stopping patience 12, 30% expanded annotation-derived ROI crops, 224-pixel EfficientFormer input, backbone/head learning rates of 5×10^{-5} and 1.5×10^{-4}, weight decay 10^{-4}, dropout 0.3, EMA, horizontal-flip TTA, temperature scaling, and top-3 checkpoint averaging. The EfficientFormer-L1+ECA variant adds a lightweight efficient-channel-attention refinement to the student feature stream before the binary head. Noisy or detector-derived ROIs are used for robustness evaluation, not for retuning the KD loss. The post-processing module fixes the calibration temperature, decision threshold, optional TTA/checkpoint-averaging choice, and ONNX Runtime export path before test reporting. Android deployment uses the same TN5000 ROI preprocessing and validation-selected post-processing logic used by the desktop reports.

4. Datasets and experimental protocol

4.1. Datasets

Table 1 summarizes the public datasets and split protocols. TN5000 uses the official train/validation/test split. BUSI and AUL use a fixed held-out test set and five training folds on the remaining trainval set; their mean and standard deviation therefore reflect training-fold variation evaluated on the same fixed test set, not an outer five-fold cross-validation estimate. BUSI is kept in its public benchmark form for comparability, but we additionally audit duplicate and near-duplicate candidates because prior work has reported BUSI curation issues [16]. The audit is reported in the supplementary material and is not used to silently change the main benchmark denominator.

4.2. Training, selection, and reporting

Models are trained with the same ROI construction, label mapping, metric computation, validation-derived threshold search, temperature-scaling procedure [21], and test-time augmentation convention unless an exception

Table 1: Datasets and protocol. Benign/malignant counts follow the descriptive counts used for the dataset overview; BUSI and AUL are used in a benign–malignant lesion-classification setting.

Dataset	Organ	Images	B/M	Split protocol
TN5000	Thyroid	5000	1426/3574	Official 3500/500/1000 train/val/test
BUSI	Breast	647	437/210	Held-out test $n = 129$; 5 folds on trainval $n = 518$
AUL	Liver	635	200/435	Held-out test $n = 127$; 5 folds on trainval $n = 508$

is stated. Timm baselines use ImageNet-pretrained initialization where available and replace their classification heads with binary heads; custom TransXNet-family models use the same local training protocol and the 1000-D task-projection head described above. Architecture choices, calibration temperatures, and operating thresholds are selected on validation predictions. Test predictions are generated after each configuration is fixed.

The detector-derived ROI condition uses diagnosis-agnostic one-class YOLO11n lesion detectors trained only to localize lesion boxes. TN5000 detectors are trained on the official train split for three seeds, with validation on the official validation split and final localization/classification evaluation on the official test split. BUSI and AUL use one detector per training fold and evaluate detector boxes on the fixed held-out test set. Detector training uses 80 epochs, 640-pixel detector input, the best validation checkpoint, confidence threshold 0.001 at inference, IoU/NMS threshold 0.7 for the TN5000 pipeline, and the highest-confidence predicted box. If no box is returned, the auto-ROI evaluator records a no-detection case and uses the uncropped source-image extent as a fallback crop; this fallback is not reported as a separate control condition. Detector weights are used only as an auxiliary front end; they are not presented as a detection-method contribution.

The main metrics are AUC, balanced accuracy (BalAcc), macro F1, accuracy, sensitivity, specificity, PPV, NPV, calibration error, and Brier score. All reported experiments use fixed split files, validation-derived model selection, and configuration-specific test reports. Because BUSI/AUL reuse a fixed held-out test set across five training folds and TN5000 repeated runs use the same official test set, the seed/fold standard deviations in benchmark tables quantify training-run variability on a fixed test population rather than independent outer-fold uncertainty. Strong ImageNet initialization, EMA, checkpoint averaging, and TTA can further reduce repeated-run AUC variance; these standard deviations should therefore be read as run-stability summaries, not as independent patient-level confidence intervals. The public release package is organized around aggregate result CSVs, figure-generation scripts, validation scripts, split/protocol metadata, and environment notes; full per-case prediction files and image-derived artifacts are retained outside the lightweight public repository when restricted by dataset or storage constraints.

4.3. Compared protocols

The experiments instantiate the four UltraDrill modules. First, the dataset-specific benchmark establishes the clean-ROI accuracy context for preprocessing and teacher candidates. Second, full-factorial TN5000 ablation identifies a robust high-capacity TransXNet-family teacher. Third, the localization-stress protocol evaluates both teacher and student models under GT boxes, synthetic box noise, detector boxes, and full-image input. Fourth, exported students are measured on two Android devices under the fixed post-processing and runtime protocol. A final multi-organ study compares joint multi-organ training with leave-one-domain-out (LODO) transfer to define the boundary of public-dataset generalization. This structure is designed to prevent three common interpretation errors: treating multi-dataset in-domain testing as zero-shot generalization, treating parameter efficiency as mobile speed, and treating oracle ROI classification as a complete clinical workflow. Table 2 defines the names used across these blocks.

Table 2: Roles of the reported model configurations inside UltraDrill. The framework reports a teacher design space and a distilled edge student rather than a single model that is optimal for every protocol.

Name	Structure	Primary role	Main evidence
UL-TransXNet	MCA+MUDD+DA teacher	Dataset-specific benchmark	Tables 3–4
TransXNet-MUDD+DA	MUDD+DA teacher	Robustness-selected teacher	Tables 5–6
EfficientFormer-L1+KD	Distilled student	Student robustness comparison	Tables 6–7
EfficientFormer-L1+ECA+KD	Distilled ECA student	Final mobile candidate	Tables 6–7
EfficientFormer-L1	Lightweight shared model	Joint/LODO multi-organ probe	Table 8

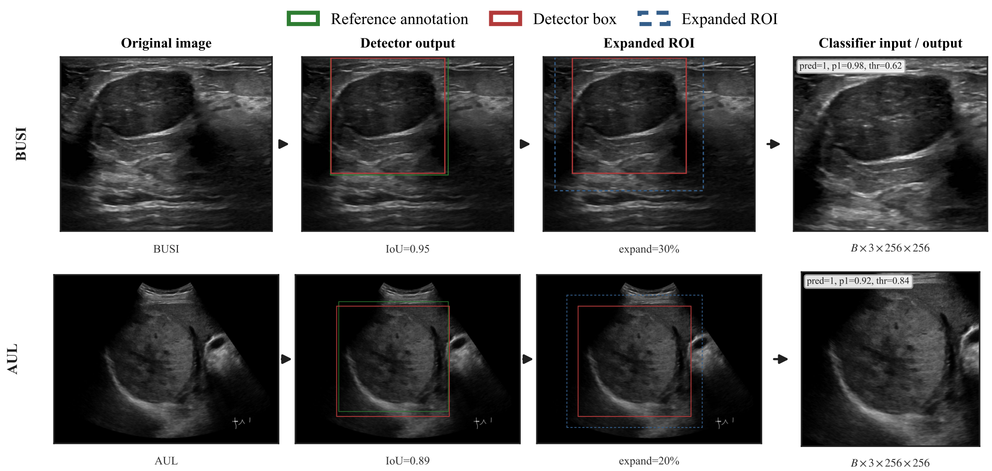

Figure 3: Automatic ROI workflow used to connect the controlled oracle-ROI classifier to a detector-classifier inference setting. Green boxes indicate reference annotations used for evaluation; red boxes indicate detector-predicted lesion boxes; blue dashed boxes indicate expanded classifier ROIs.

5. Results

5.1. Dataset-specific benchmark and model-family context

Table 3 summarizes the dataset-specific benchmark results that motivate the later robustness and deployment analyses. The result should be read as multi-organ public benchmark evidence under dataset-specific training, not as zero-shot transfer. Under the fixed reporting protocol, UL-TransXNet is slightly stronger than ConvNeXt-Tiny on TN5000 and stronger than Swin-T on BUSI. On AUL, however, Swin-T remains stronger by AUC and balanced accuracy, so we do not claim uniform dataset-level superiority.

Table 3: Main dataset-specific benchmark summary. Values are mean ± standard deviation over TN5000 seeds or BUSI/AUL training folds evaluated on the fixed held-out test sets. F1 denotes macro F1.

Dataset	Method	AUC	BalAcc	F1	Acc
TN5000	UL-TransXNet	**0.9537 ± 0.0015**	**0.8865 ± 0.0064**	**0.8750 ± 0.0059**	**0.8960 ± 0.0075**
TN5000	ConvNeXt-Tiny	0.9508 ± 0.0020	0.8794 ± 0.0023	0.8731 ± 0.0066	0.8957 ± 0.0065
BUSI	UL-TransXNet	**0.8494 ± 0.0039**	**0.7702 ± 0.0076**	**0.7479 ± 0.0283**	**0.7752 ± 0.0383**
BUSI	Swin-T	0.8190 ± 0.0091	0.7137 ± 0.0095	0.6896 ± 0.0300	0.7194 ± 0.0445
AUL	UL-TransXNet	0.8146 ± 0.0165	0.7611 ± 0.0283	**0.7589 ± 0.0307**	**0.7827 ± 0.0329**
AUL	Swin-T	**0.8319 ± 0.0117**	**0.7783 ± 0.0325**	**0.7589 ± 0.0243**	0.7717 ± 0.0211

Because the method is built from the TransXNet family, Table 4 reports the direct comparison with the original TransXNet under the same dataset-specific reporting protocol. UL-TransXNet improves AUC and balanced accuracy over the original TransXNet on all three datasets, but the size of the gain is dataset-dependent and does not make the model the best baseline on AUL.

Table 4: Original TransXNet versus UL-TransXNet under the same dataset-specific reporting protocol. Δ denotes UL-TransXNet minus TransXNet.

Dataset	TransXNet		UL-TransXNet		Δ	
	AUC	BalAcc	AUC	BalAcc	AUC	BalAcc
TN5000	0.9467	0.8727	0.9537	0.8865	+0.0070	+0.0138
BUSI	0.8136	0.7531	0.8494	0.7702	+0.0358	+0.0171
AUL	0.7899	0.7566	0.8146	0.7611	+0.0247	+0.0045

5.2. TransXNet-family ablation

Table 5 reports the TN5000 full-factorial ablation under the fixed reporting protocol. The results clarify that performance does not come from simply stacking every available module. The strongest mean AUC is obtained by combining MUDD and differential attention without MCA, reaching AUC 0.9579 and BalAcc 0.8901. MCA improves some operating behavior but is not uniformly beneficial: the full MCA+MUDD+DA variant reaches AUC 0.9522 and BalAcc 0.8754 in this sweep. We therefore describe MCA as a dataset-dependent refinement rather than a mandatory part of the final claim.

5.3. Localization-stress transfer from teacher to student

Table 6 applies the same TN5000 localization-stress protocol to the two TransXNet-family teachers and the three EfficientFormer-L1 students. This table is the main bridge between teacher analysis and mobile distillation. The MUDD+DA teacher remains the best clean-ROI model in this block (GT AUC 0.9623), but it loses 0.0160 AUC under 20% box noise. UL-TransXNet loses 0.0224 AUC under the same perturbation. In contrast, the KD students preserve most of their clean-ROI performance: EfficientFormer-L1+KD obtains GT AUC 0.9585, 20% box-noise AUC

Table 5: Full-factorial TN5000 ablation of the TransXNet-family teacher. Values are mean test metrics over three seeds under the fixed reporting protocol.

Variant	MCA	MUDD	DA	AUC	BalAcc
TransXNet-base	0	0	0	0.9456	0.8769
TransXNet-DA	0	0	1	0.9447	0.8785
TransXNet-MUDD	0	1	0	0.9511	0.8857
TransXNet-MUDD+DA	0	1	1	**0.9579**	**0.8901**
TransXNet-MCA	1	0	0	0.9465	0.8802
TransXNet-MCA+DA	1	0	1	0.9441	0.8747
TransXNet-MCA+MUDD	1	1	0	0.9555	0.8834
UL-TransXNet	1	1	1	0.9522	0.8754

0.9529, and detector-ROI AUC 0.9534; EfficientFormer-L1+ECA+KD obtains GT AUC 0.9557, 20% box-noise AUC 0.9513, and detector-ROI AUC 0.9523. The non-KD EfficientFormer-L1+ECA student is substantially weaker on clean ROIs, showing that the retained performance is not merely a property of the lightweight backbone. Full-image AUCs remain far below ROI-based AUCs, confirming that the pipeline still depends on lesion localization.

Table 6: TN5000 teacher-student localization-stress test. All rows use the same validation-derived operating point per model. Drop@20% is GT-ROI AUC minus 20% box-noise AUC.

Model	GT ROI AUC	20% noise AUC	Drop@20%	Detector ROI AUC	Full-image AUC
UL-TransXNet teacher	0.9600	0.9376	0.0224	0.9487	0.6556
TransXNet-MUDD+DA teacher	**0.9623**	0.9463	0.0160	0.9497	0.6869
EfficientFormer-L1+ECA student	0.9280	0.9258	**0.0021**	0.9308	0.7165
EfficientFormer-L1+KD student	0.9585	**0.9529**	0.0056	**0.9534**	0.7063
EfficientFormer-L1+ECA+KD student	0.9557	0.9513	0.0043	0.9523	**0.7111**

5.4. Edge post-processing and mobile feasibility

Table 7 reports the mobile results after re-export and two-device remeasurement. Offline ensemble metrics are taken from the strict mobile model-level summary, while Android AUC is computed from the exported single ONNX artifact under the same TN5000 test protocol. The final mobile candidate is EfficientFormer-L1+ECA+KD, which reaches 0.9568 offline ensemble AUC, 0.9552 Android-exported AUC, and 31.3/56.5 ms hot end-to-end latency on the Xiaomi phone and Samsung tablet. The runtime sweep also

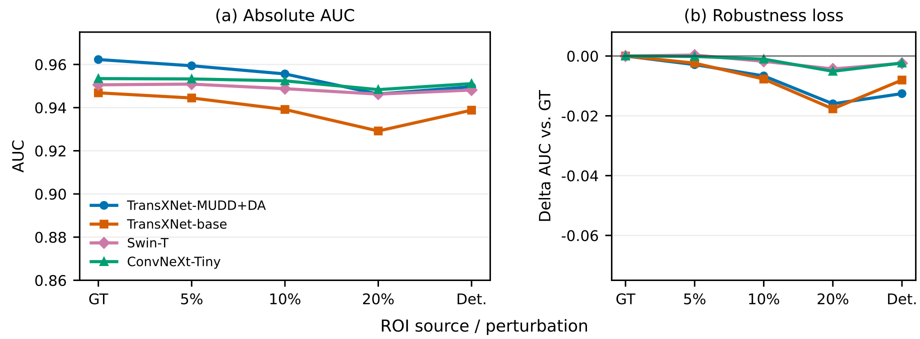

Figure 4: Teacher-side AUC robustness under GT boxes, synthetic perturbations, and detector-derived boxes. The left panel compares absolute AUC; the right panel shows degradation from each model's GT-ROI AUC.

shows that the default CPU backend is faster than XNNPACK and NNAPI on both measured devices for this model.

Table 7: TN5000 mobile feasibility under the fixed reporting protocol. Offline metrics and Android-exported AUC use the same TN5000 test set; latency is hot end-to-end latency from exported artifacts on a Xiaomi 24129PN74C phone and Samsung SM-X800 tablet.

Model	Offline AUC	Offline BalAcc	Android AUC	Xiaomi ms	Samsung ms
EfficientFormer-L1+ECA+KD	0.9568	**0.8816**	**0.9552**	**31.312**	56.507
EfficientFormer-L1+KD	**0.9586**	0.8781	0.9511	40.364	**56.477**
EfficientFormer-L1+ECA	0.9270	0.8441	0.9278	40.517	60.254

5.5. Multi-organ joint training and transfer boundary

Table 8 separates joint multi-organ training from true held-out-organ transfer. Joint training trains one shared EfficientFormer-L1 model and evaluates each dataset under its own test protocol. LODO tests whether a model trained without the target organ transfers directly to that held-out organ. Joint training remains feasible, but LODO AUCs of 0.5936–0.7339 show that zero-shot cross-organ generalization is weak. We therefore use the term multi-organ public benchmark evaluation, not cross-organ validation.

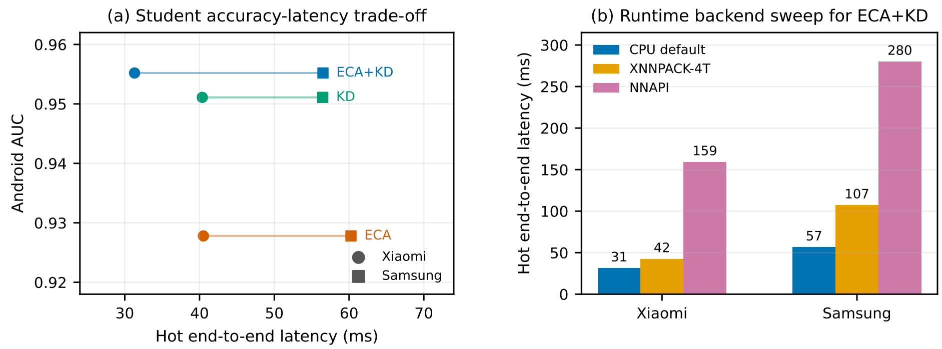

Figure 5: Two-device Android feasibility summary. (a) The CPU-default exported students show the accuracy–latency trade-off on Xiaomi and Samsung devices. (b) For the final EfficientFormer-L1+ECA+KD student, the runtime sweep shows that CPU execution is faster than XNNPACK-4T and NNAPI on both measured devices.

Table 8: Multi-organ joint training and zero-shot boundary probe with EfficientFormer-L1. Values are mean ± standard deviation over three seeds.

Protocol	TN5000 AUC	BUSI AUC	AUL AUC
Joint multi-organ	0.9212 ± 0.0065	0.8324 ± 0.0192	0.7984 ± 0.0367
LODO zero-shot	0.6261 ± 0.0557	0.7339 ± 0.0428	0.5936 ± 0.0344

6. Discussion

The central result is not that one backbone is uniformly best. It is that UltraDrill turns localization stress into a practical optimization criterion for edge ultrasound classification. Clean-ROI benchmarking identifies high-capacity candidates, full-factorial ablation shows which TransXNet-family teacher is most reliable, the same perturbation protocol checks whether a compact student preserves performance when lesion boxes are imperfect, and fixed post-processing/export makes the mobile claim measurable. This sequence is more useful than reporting a teacher and a mobile model as unrelated experiments, because it asks whether deployment compression changes the model's sensitivity to the very ROI error that motivated the framework.

The factorial ablation also changes the interpretation of the modules. MUDD and differential attention provide the most reliable TN5000 teacher gain. MCA is not a universal AUC booster, even though it may still help some operating points and datasets. We therefore treat the TransXNet-family model as a teacher design space, not as a monotonic sequence where every added module improves every metric. This distinction is important for the paper's claim: the teacher is selected for analysis and distillation, not presented as a universally superior backbone.

The matched teacher-student localization-stress comparison is the strongest practical evidence for UltraDrill. The MUDD+DA teacher gives the best clean-ROI AUC in the robustness block, but the distilled EfficientFormer-L1 students show smaller AUC degradation under 20% synthetic box noise and detector-derived boxes. This does not prove that students are inherently more robust; lower-capacity models can also be less sensitive to crop perturbations because they rely on coarser features. The relevant finding is narrower and stronger: distillation did not amplify localization sensitivity, and the exported student remains close to the teacher under detector-derived ROI inputs. Full-image AUC remains much lower than ROI-based AUC, so the framework still requires a lesion-localization step.

The Android experiment then completes the UltraDrill framework rather than serving as a separate demo. The student selected by accuracy, robustness, and exportability reaches 0.9552 Android AUC with 31.3/56.5 ms hot latency on the Xiaomi phone and Samsung tablet. Runtime profile sweeps showed that CPU execution was faster than XNNPACK and NNAPI for the exported EfficientFormer-L1+ECA+KD artifact on both devices. This reinforces the reason to separate teacher and student roles: a teacher can be valuable for robust representation analysis without being the right endpoint for mobile inference.

The multi-organ results define a second boundary. BUSI and AUL are useful public in-domain benchmarks from different organs, but LODO performance remains weak. The framework should therefore be read as a mobile ROI-classification workflow evaluated across multiple public datasets, not as a universal cross-organ ultrasound classifier. Calibration remains a separate constraint from discrimination: the current manuscript focuses on discrimination, localization robustness, and Android latency, while probability outputs and operating thresholds require site-specific validation before clinical use.

Limitations

This study has several limitations. First, all datasets are retrospective public datasets, and the label definitions and acquisition conditions are not fully harmonized. The BUSI audit identifies exact and perceptual duplicate candidates, including label-conflict candidates, so BUSI results should be interpreted as public-benchmark comparability rather than a cleaned patient-level clinical estimate. Second, annotation-derived ROI crops remain the primary classifier protocol; detector-derived crops and box perturbations mitigate but do not eliminate this dependency. Third, BUSI and AUL are primarily dataset-specific benchmarks, not evidence of thyroid-to-breast or thyroid-to-liver external generalization. Fourth, mobile feasibility is measured on two Android devices and should be interpreted as prototype evidence until broader chipsets, delegates, and thermal states are measured. Fifth, probability calibration and site-specific operating thresholds require further per-case validation under the same fixed reporting discipline used for the discrimination results. Sixth, the module-level changes are adaptations of existing ideas; the contribution is the systematic UltraDrill framework that links ROI preprocessing, robustness-aware teacher optimization, student distillation, and edge verification.

7. Conclusions

We presented UltraDrill, a distillation and optimization framework for edge ultrasound lesion classification. The key idea is to make ROI perturbation a framework-level selection and verification tool: inputs are preprocessed and stress-tested, high-capacity TransXNet-family teachers are optimized and selected under controlled localization errors, EfficientFormer students are distilled and tested under the same stress conditions, and fixed post-processing/export is verified on Android devices. On TN5000, the MUDD+DA teacher provides the strongest clean-ROI teacher evidence, while KD students preserve high AUC under 20% box noise and detector-derived boxes and run at practical two-device CPU latency after export. The result is a bounded edge-classification framework, not a claim of clinical deployment readiness or universal cross-organ generalization.

CRediT authorship contribution statement

Zihan Li: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft.
Hongyu Wei: Clinical interpretation, Medical domain review, Validation,

Writing – review and editing.
Cong Yang: Conceptualization, Supervision, Resources, Project administration, Writing – review and editing.

Ethics statement

This study is a secondary computational analysis of publicly available, de-identified ultrasound datasets. No new patient data were collected by the authors. Ethical approval and informed-consent procedures are governed by the original dataset publications or data providers; therefore, no additional institutional review board approval was required for this retrospective computational study.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used ChatGPT to support language editing, organization of revision evidence, and consistency checks under human oversight. The authors reviewed and edited all content and take full responsibility for the final manuscript.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed in this study are publicly available from their original sources, which are cited in the manuscript. A reproducibility package is prepared for release and includes source code, split/protocol metadata, aggregate table CSVs, figure-generation scripts, validation scripts, environment notes, Android export hashes, and two-device runtime summaries. Dataset image files, generated ROI folders, trained checkpoints, detector weights, ONNX binaries, Android packages, full raw logs, and full per-case prediction files are not redistributed in the lightweight repository; they may be made available from the corresponding author upon reasonable request where permitted by dataset, privacy, storage, and software terms.

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